Project Requirements — Lights Out Solver

Project name: Lights Out Solver

Revision: 0.1

Status: Initial draft

1. Objective

Design and build a physical device that accepts a 5 x 5 Lights Out puzzle, calculates a valid solution, and displays the grid positions the user must press to solve the puzzle>

2. Functional requirements

- FR-01: The device shall accept the initial state of all 25 puzzle cells.
- FR-02: The device shall represent each cell as either ON or OFF.
- FR-03: The device shall calculate a solution using a linear-algebra-based algorithm.
- FR-04: The device shall display the recommended moves on a 5 x 5 LED grid.
- FR-05: The device shall indicate when a puzzle has no valid solution.
- FR-06: The device shall provide a way to reset the puzzle and enter another one.

3. Design constraints

- The display shall contain 25 LED positions arranged in a 5 x 5 grid.
- The software shall be written in C++.
- The design shall include an electronic schematic and PCB layout.
- The enclosure and component placement shall be documented using CAD.
- The device shall operate using an appropriate, low-voltage power supply.

4. Verification criteria

- The solver shall be tested against known puzzle configurations, including unsolvable configurations.
- Every reported solution shall be checked by applying its moves to a copy of the original state.
- The physical LED display shall match the calculated solution.

- The reset and puzzle-entry functions shall be tested.

5. Open design decisions

- Methods of puzzle entry
- Controller and LED driver selection
- LED color and display modes
- Power source
- Whether a PLC is part of the final hardware architecture
- Enclosure dimensions and fabrication method